f. Run APPFabric Applications LOCALLY with Eclipse IDE

The APPFabric Engineering "Getting Started" Pages do a great job of describing how to run the Proxy, Web, & API Applications *LOCALLY* to your developer machine; however they have done so while leveraging the "**IntelliJ IDE**".

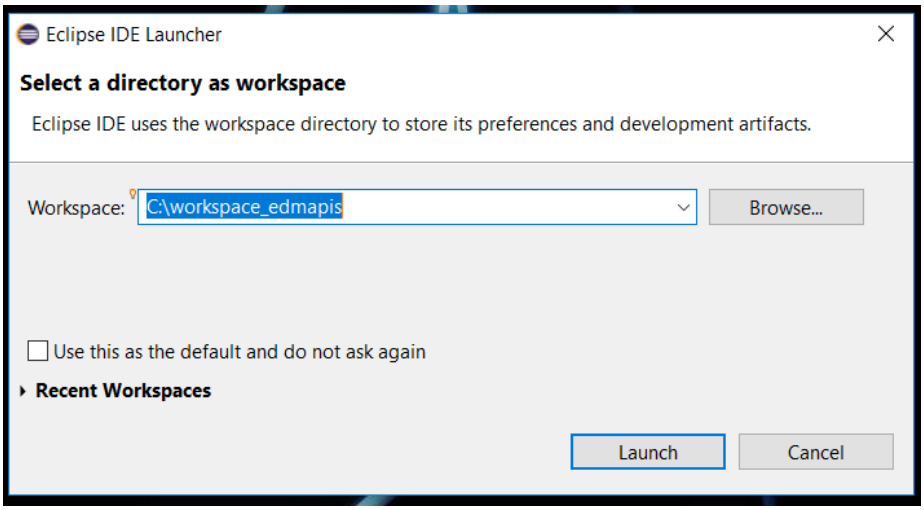
(Reference the following webpage; https://github.allstate.com/pages/Engineering/AppFabric/docs/gettingStartedRunLocally.html).

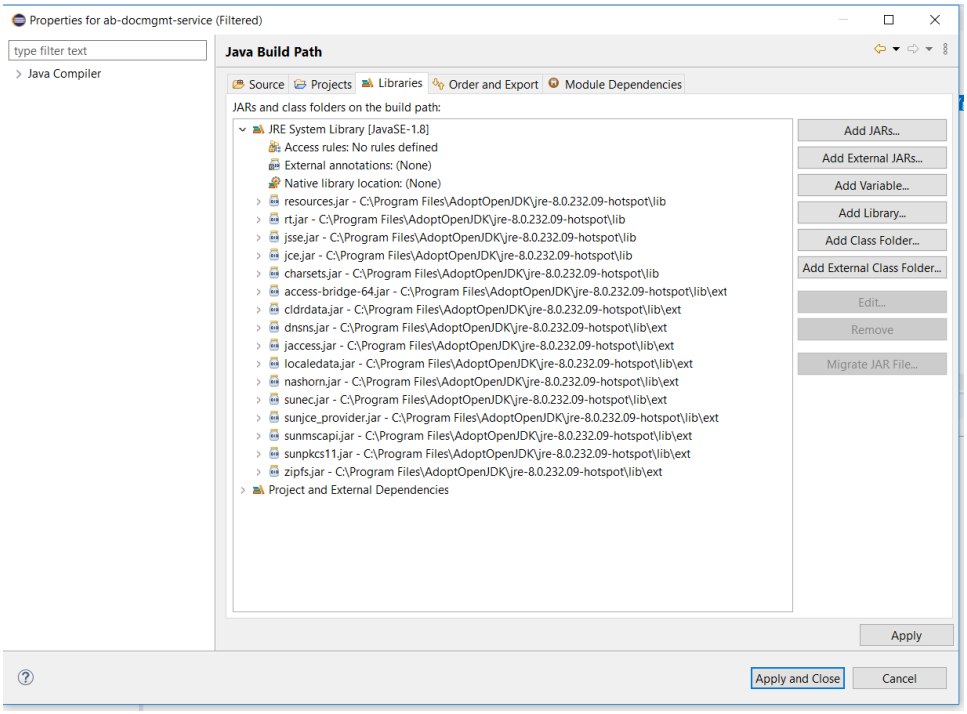
We have been going down the path of leveraging an **ECLIPSE IDE** NOT **IntelliJ IDE.**

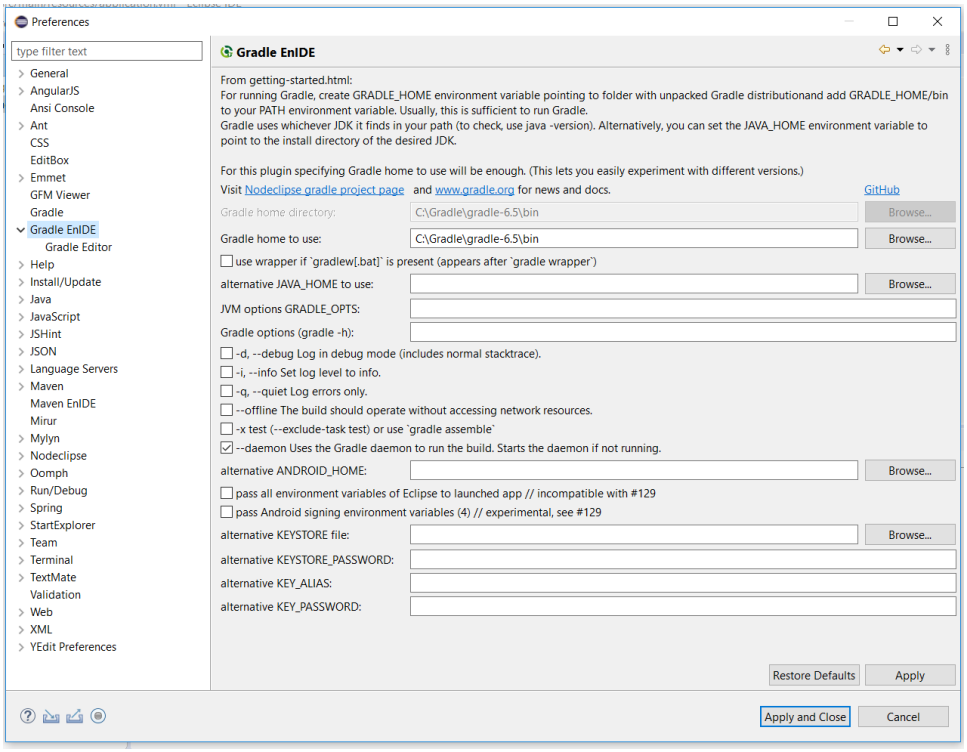
In so doing, it was felt that there should be an additional set of instructions dedicated to **ECLIPSE**; which is the intent of this page.

> **NOTE:** This list of instructions has been published for the **Eclipse LUNA IDE**.

> **NOTE:** For everything to work, the proxy and API must be run with the local Spring profile. Profiles are found within the designated application's "**application.yml**" File
>
> The developer may reference the following links for a more indepth understanding of SpringBoot, Eclipse, and default Tomcat Server Configurations:
>
> https://www.baeldung.com/eclipse-sts-spring
>
> https://www.baeldung.com/spring-yaml
>
> https://reflectoring.io/spring-boot-profiles/
>
> https://www.baeldung.com/spring-boot-change-port
>
> https://www.java67.com/2019/07/spring-boot-3-ways-to-change-port-of-tomcat.html

Step	Instructions	Description / Comments
1.	Open Eclipse	Navigate to your Applications Workspace. Click the "**Launch**" Button 
2	If you have not already done so Import your projects	Follow the Steps found on our Team Confluence Page: Importing an Existing Project & Build from GitHub

3	Validate that the Project JAVA JRE's have been configured for JAVA_HOME	Navigate to the following: Right-Click on the Project Build Path Configure Build Path The following Child Window will appear (Reference Below) Click the Libraries Tab and open the JRE System Instance. Valid Configuring Your Developer IDE for NODE.js, JAVA 8, and other Dependencies 

4	Validate that Gradle has been configured for GRADLE_HOME	Navigate to the following: Windows Preferences GradleEnIDE The following Child Window will appear (Reference Below) Click the Libraries Tab and open the JRE System Instance. Valid Configuring Your Developer IDE for NODE.js, JAVA 8, and other Dependencies 

5	Familiarize yourself with the Application's YAML Files "application.yml"	These files contain the application specific **SpringBoot** Profile Attributes. For example our newly generated "**Service**" Project
		
		This is critical information. It will prove vital when we configure our "**Service**" Project "**Run Configurations**"
		
		build.gradle *application.yml application.yml
		
		``yaml
		--- 	
		spring:
		  profiles: local
		# The API is secured with a JWT token for local profile. For further details
		# Please see the following
		# https://github.allstate.com/pages/Engineering/AppFabric/docs/gettingStartedRunLocally.html
		
		server:
		  port: 8081
		
		``
		
		Snapshot of the Proxy Project **SpringBoot** Profiles for reference:

build.gradle | *application.yml | *application.yml

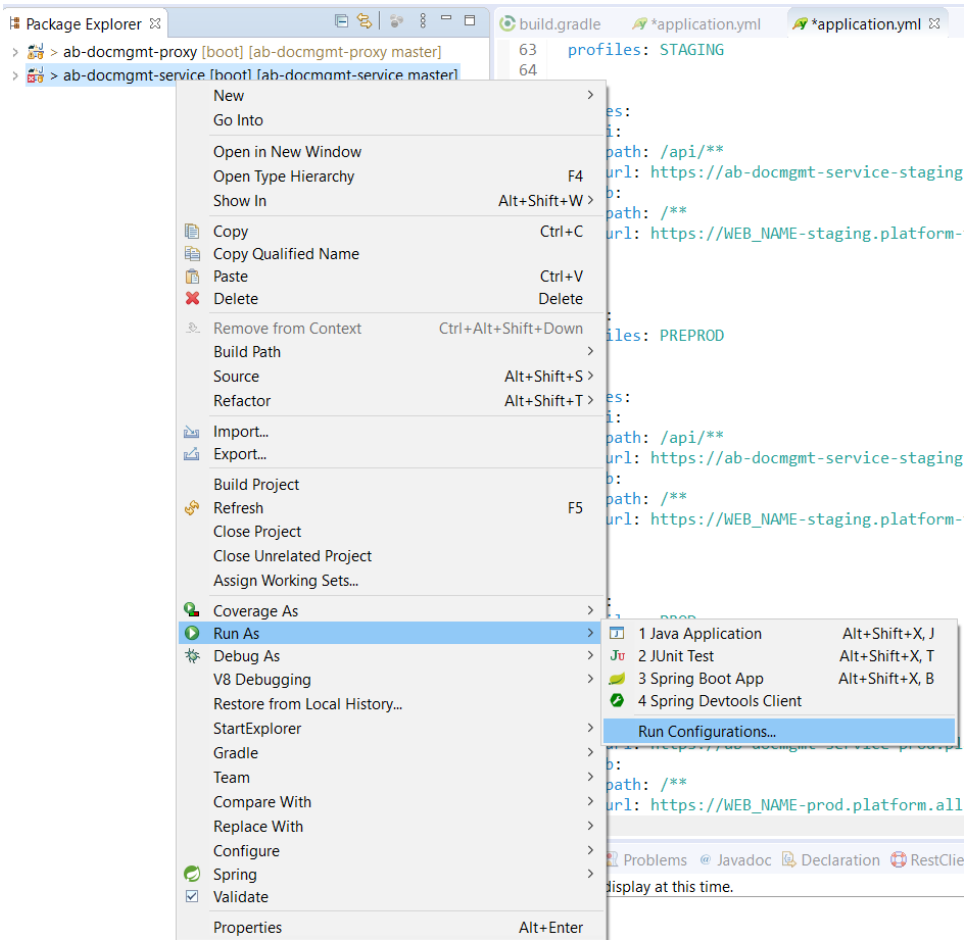
```yaml
---

spring:
  profiles: local
# The API is secured with a JWT token for local profile.
# For further details, please see the following:
# https://github.allstate.com/pages/Engineering/AppFabric/docs/gettingStartedRunLocally.html

zuul:
  routes:
    api:
      path: /api/**
      url: http://localhost:8081
    web:
      path: /**
      url: http://localhost:9000

---

spring:
  profiles: DEV

zuul:
  routes:
    api:
      path: /api/**
      url: https://ab-docmgmt-service-dev.platform-test.allstate.com
    web:
      path: /**
      url: https://WEB_NAME-dev.platform-test.allstate.com

---

spring:
  profiles: UAT

zuul:
  routes:
    api:
      path: /api/**
      url: https://ab-docmgmt-service-uat.platform-test.allstate.com
    web:
      path: /**
      url: https://WEB_NAME-uat.platform-test.allstate.com

---

spring:
  profiles: INT

zuul:
  routes:
    api:
      path: /api/**
      url: https://ab-docmgmt-service-int.platform-test.allstate.com
    web:
      path: /**
      url: https://WEB_NAME-int.platform-test.allstate.com

---

spring:
  profiles: STAGING

zuul:
  routes:
    api:
      path: /api/**
      url: https://ab-docmgmt-service-staging.platform-test.allstate.com
    web:
      path: /**
      url: https://WEB_NAME-staging.platform-test.allstate.com

---

spring:
  profiles: PREPROD

zuul:
  routes:
    api:
      path: /api/**
      url: https://ab-docmgmt-service-staging.platform-test.allstate.com
    web:
      path: /**
      url: https://WEB_NAME-staging.platform-test.allstate.com

---
```

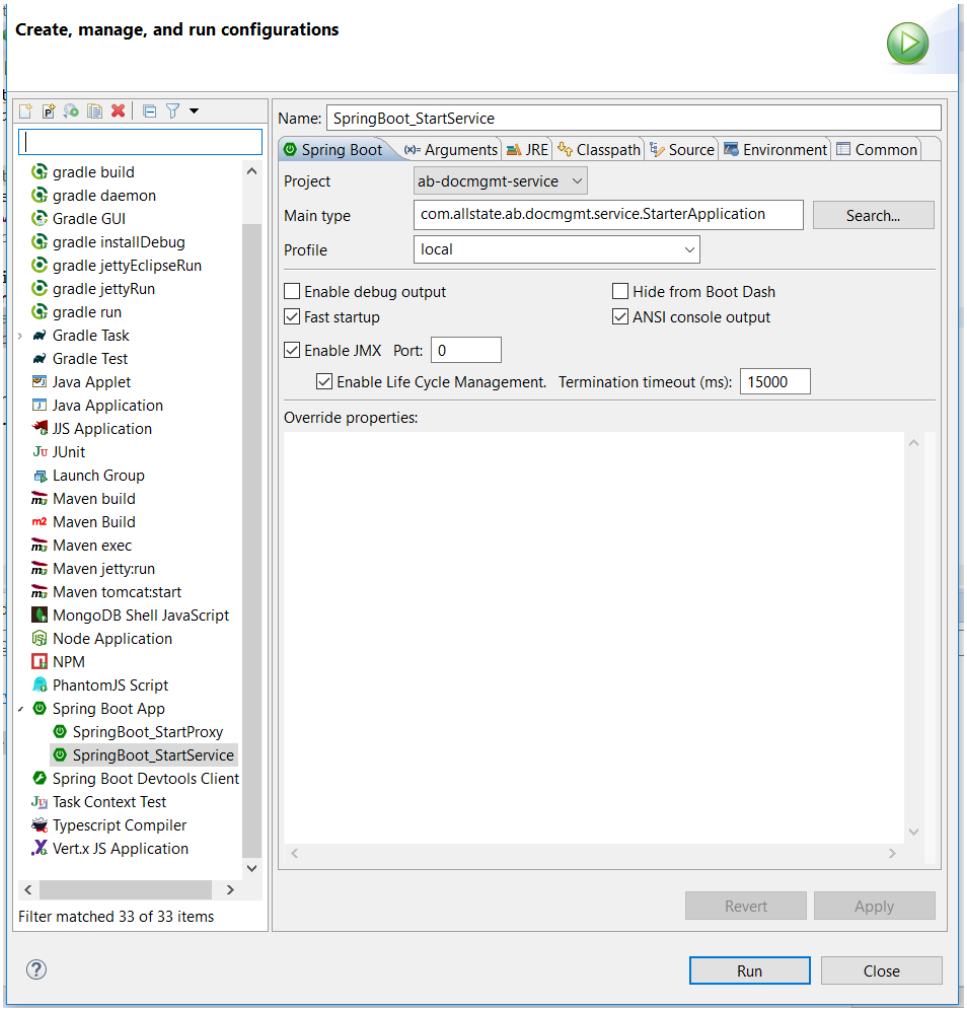
```
89
90 spring:
91   profiles: PROD
92
93 zuul:
94   routes:
95     api:
96       path: /api/**
97       url: https://ab-docmgmt-service-prod.platform.allstate.com
98     web:
99       path: /**
100      url: https://WEB_NAME-prod.platform.allstate.com
```

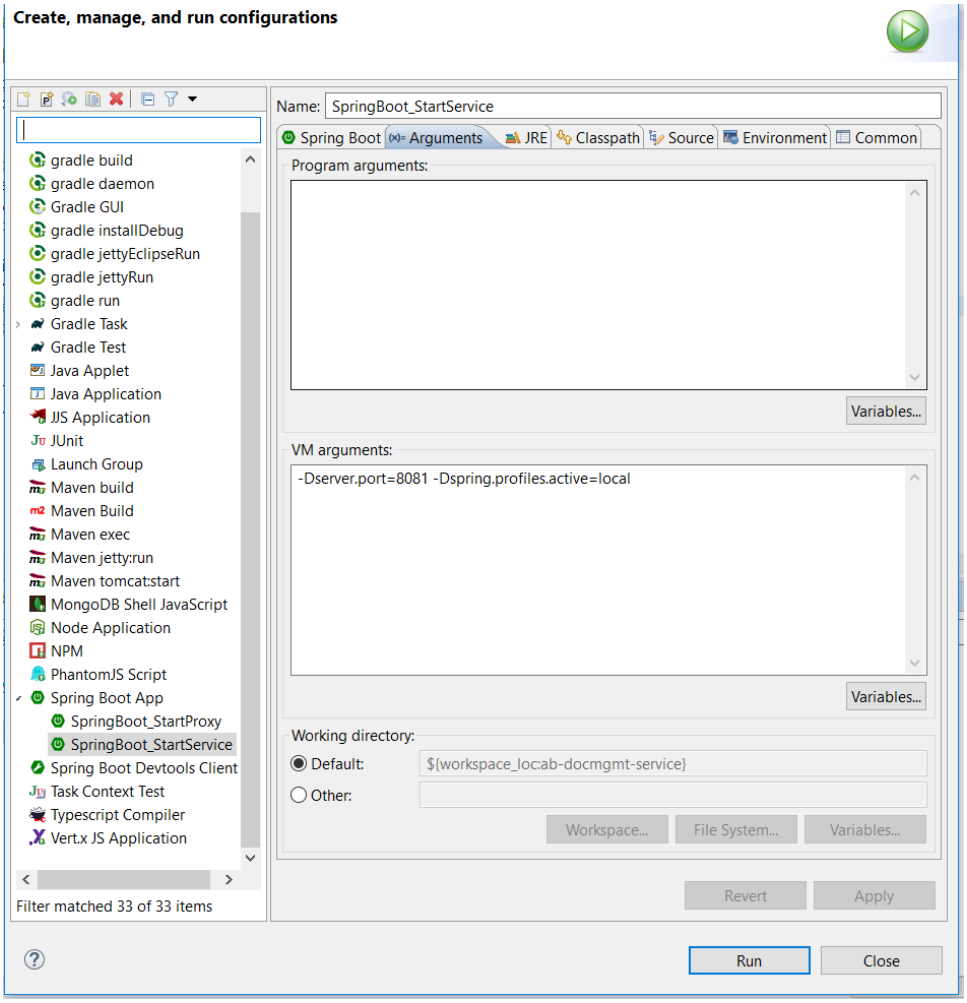
6 Configure the Project SpringBoot Run Configurations

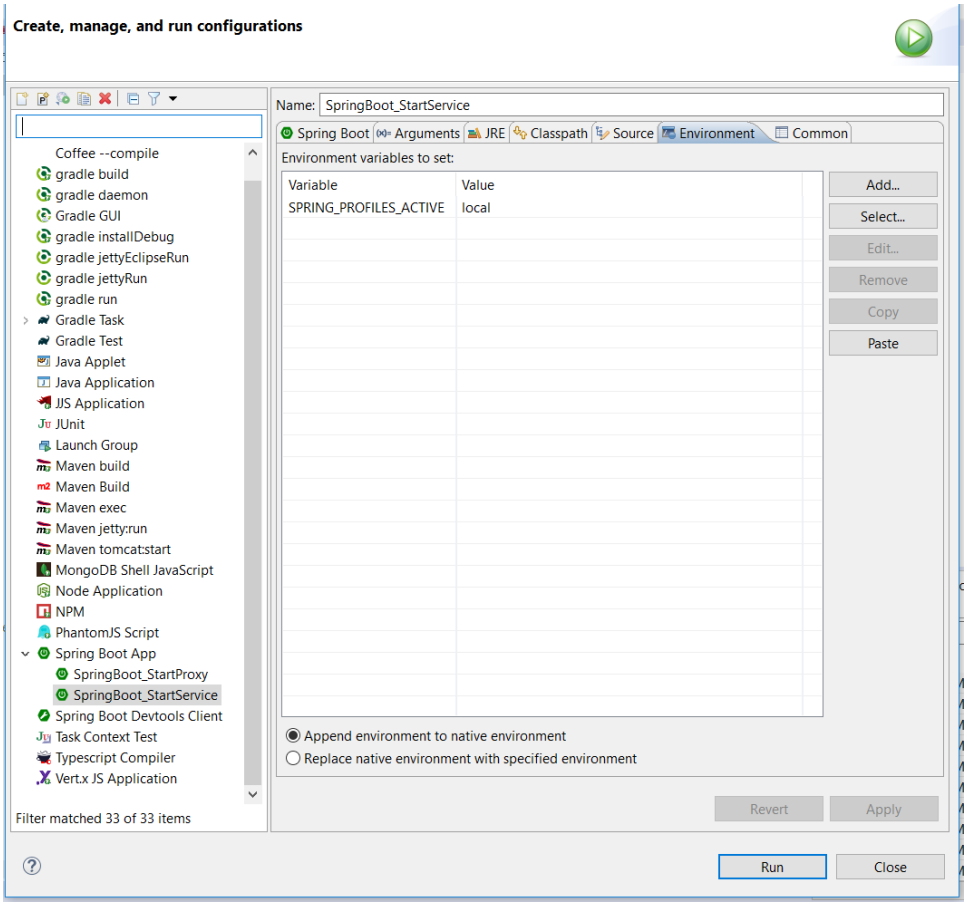
Navigation to the following:

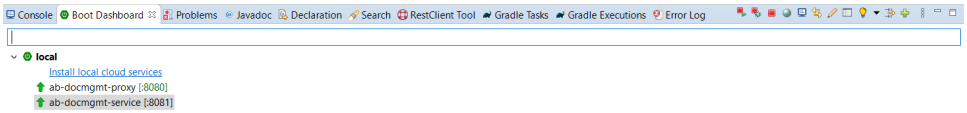
Right-click Run As Run Configurations

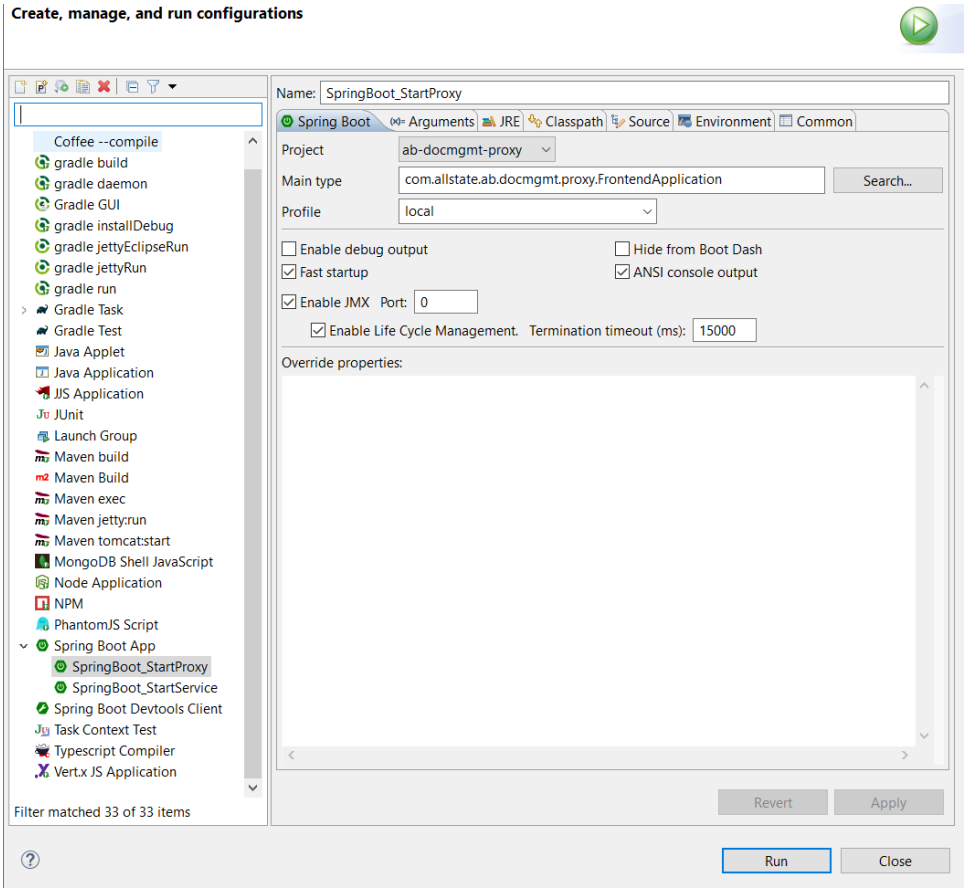


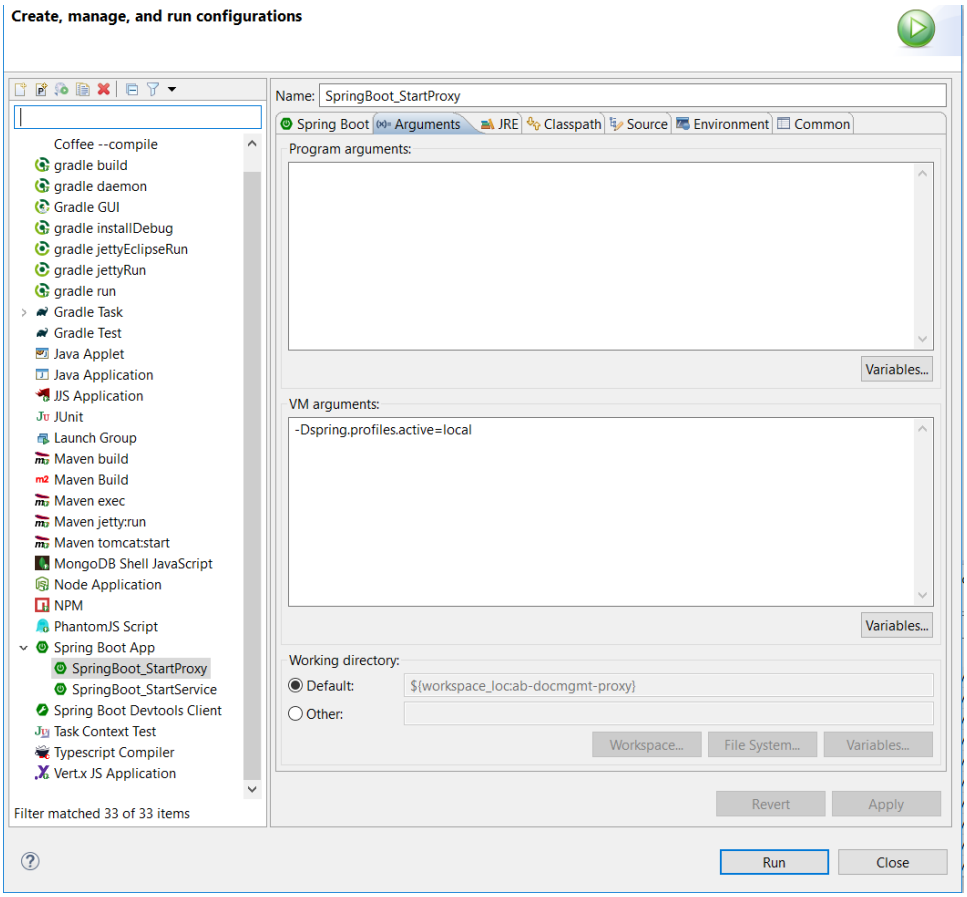
7	Create a New SpringBoot Run Application Configuration	As shown in the screen-shot below, create a new SpringBoot Run Configuration for your project by providing the following inf **a. Name:** We chose SpringBoot_StartService **b. Project:** We chose ab-docmgmt-service **c. Main Type:** This is your project application's Class that contains the "**MAIN**" Body Once provided, Click the "**Apply**" Button 

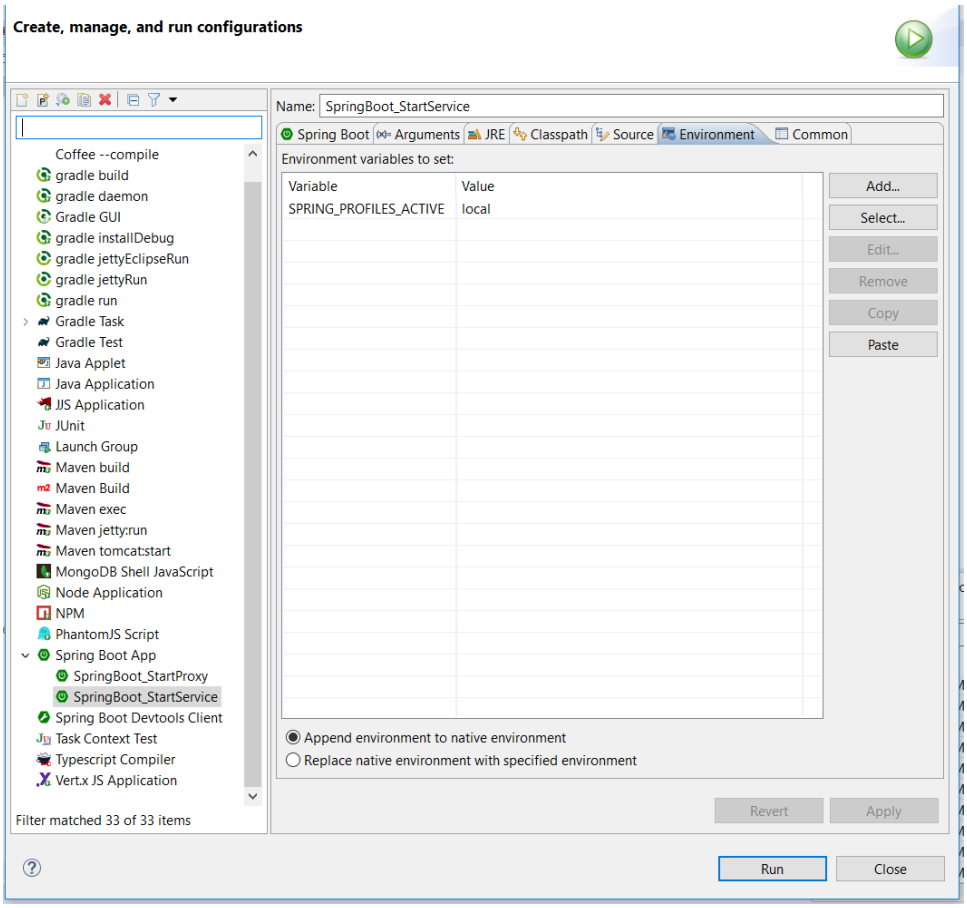
8	Click on the "**Arguments**" Tab, within the Run Configurations	Here is where familiarizing ourselves with the "**application.yml**" file becomes important. Click the "**Arguments**" Tab and Enter the **SpringBoot Profile,** from the project's **"application.yml"** file**,** as a **VM Argument** "-Dserver.port=8081 -Dspring.profiles.active=local" Once provided, Click the "**Apply**" Button 

9	Create an Environment Variable within the Configuration	Click the "**Environment**" Tab and Create an Environment Variable for the **SpringBoot Profile,** from the project's **"applicatio** **"SPRING_PROFILES_ACTIVE" : "local"** Once provided, Click the "**Apply**" Button This may seem like we are repeating **Step #8**, which technically we are. What VM Arguments are passed should ove ingested into the **Run Configurations** for this Application. 

10	Click the "**Run**" Button, within the Run Configurations, for the Service Project	If everything has been entered **Successfully**, the following should be written to the **ECLIPSE IDE Console Window:**
		
		Reference the 2nd Line from the Bottom - It states that the TomCat Server has started on port(s) 8081
		
		``
		  . ____ _ __ _ _
		 /\\ / ___'_ __ _ _(_)_ __ __ _ \ \ \ \
		(( )___ | '_ | '_|
			'_ \/ _`	\ \ \ \
				 \\/ ___)| |
								|| (_| |))))
								  ' |____| .__|_| |_|_| |___,
									/ / / /
									 =========|_|==============|___/=/_/_/_/
									 :: Spring Boot :: (v2.2.6.RELEASE)
									
									2020-06-11 12:44:29.195 INFO 6828 --- [main] c.a.a.d.service.StarterApplication : Starting StarterApplication on 247-HOCL0010714 with PID 6828 (C:\workspace_edma
									2020-06-11 12:44:29.198 INFO 6828 --- [main] c.a.a.d.service.StarterApplication : The following profiles are active: local
									2020-06-11 12:44:30.363 INFO 6828 --- [main] trationDelegate$BeanPostProcessorChecker : Bean 'spring.security-org.springframework.boot.autoconfigure.security.SecurityF
									2020-06-11 12:44:30.375 INFO 6828 --- [main] trationDelegate$BeanPostProcessorChecker : Bean 'org.springframework.security.config.annotation.configuration.ObjectPostPr
									2020-06-11 12:44:30.379 INFO 6828 --- [main] trationDelegate$BeanPostProcessorChecker : Bean 'objectPostProcessor' of type [org.springframework.security.config.annotat
									2020-06-11 12:44:30.381 INFO 6828 --- [main] trationDelegate$BeanPostProcessorChecker : Bean 'org.springframework.security.access.expression.method.DefaultMethodSecuri
									2020-06-11 12:44:30.384 INFO 6828 --- [main] trationDelegate$BeanPostProcessorChecker : Bean 'org.springframework.security.config.annotation.method.configuration.Globa
									2020-06-11 12:44:30.387 INFO 6828 --- [main] trationDelegate$BeanPostProcessorChecker : Bean 'methodSecurityMetadataSource' of type [org.springframework.security.acces
									2020-06-11 12:44:30.397 INFO 6828 --- [main] trationDelegate$BeanPostProcessorChecker : Bean 'com.composed.appfabric.logging.config.AppFabricLoggingConfiguration' of t
									2020-06-11 12:44:30.651 INFO 6828 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port(s): 8081 (http)
									2020-06-11 12:44:30.657 INFO 6828 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
									2020-06-11 12:44:30.658 INFO 6828 --- [main] org.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/9.0.33]
									2020-06-11 12:44:30.844 INFO 6828 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
									2020-06-11 12:44:30.844 INFO 6828 --- [main] o.s.web.context.ContextLoader : Root WebApplicationContext: initialization completed in 1563 ms
									2020-06-11 12:44:31.403 INFO 6828 --- [main] o.s.s.web.DefaultSecurityFilterChain : Creating filter chain: any request, [org.springframework.security.web.context.r
									2020-06-11 12:44:31.557 INFO 6828 --- [main] o.s.s.concurrent.ThreadPoolTaskExecutor : Initializing ExecutorService 'applicationTaskExecutor'
									2020-06-11 12:44:31.904 INFO 6828 --- [main] o.s.b.a.e.web.EndpointLinksResolver : Exposing 3 endpoint(s) beneath base path '/actuator'
									2020-06-11 16:44:31,958[appFabric-security]enable|2.6.2|
									2020-06-11 16:44:31,958[appFabric-logging]enable|2.6.2|
									2020-06-11 16:44:31,959[appFabric-starter-api]enable|2.6.2|
									2020-06-11 12:44:31.988 INFO 6828 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port(s): 8081 (http) with context path ''
									2020-06-11 12:44:31.990 INFO 6828 --- [main] c.a.a.d.service.StarterApplication : Started StarterApplication in 3.381 seconds (JVM running for 5.399)
									2020-06-11 12:44:34.641 INFO 6828 --- [on(1)-127.0.0.1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
									2020-06-11 12:44:34.642 INFO 6828 --- [on(1)-127.0.0.1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
									2020-06-11 12:44:34.650 INFO 6828 --- [on(1)-127.0.0.1] o.s.web.servlet.DispatcherServlet : Completed initialization in 8 ms
									``
									
									This can also be confirmed from the Boot Dashboard Perspective (Reference Snapshot below...)
									
									 

11	Create a New SpringBoot Run Application Configuration for the Proxy Project	As shown in the screen-shot below, create a new SpringBoot Run Configuration for your project by providing the following inf **a. Name:** We chose SpringBoot_StartProxy **b. Project:** We chose ab-docmgmt-proxy **c. Main Type:** This is your project application's Class that contains the "**MAIN**" Body Once provided, Click the "**Apply**" Button 

12	Click on the "**Arguments**" Tab, within the Run Configurations	Here is where familiarizing ourselves with the "**application.yml**" file becomes important. Click the "**Arguments**" Tab and Enter the **SpringBoot Profile,** from the project's **"application.yml"** file**,** as a **VM Argument** "-Dspring.profiles.active=local" Notice we have not configured the port, we wish for it to run on the default port of 8080 Once provided, Click the "**Apply**" Button 

13	Create an Environment Variable within the Configuration	Click the "**Environment**" Tab and Create an Environment Variable for the **SpringBoot Profile,** from the project's **"applicatio** **"SPRING_PROFILES_ACTIVE" : "local"** Once provided, Click the "**Apply**" Button This may seem like we are repeating **Step #8**, which technically we are. What VM Arguments are passed should ove ingested into the **Run Configurations** for this Application. 

14	Click the "**Run**" Button, within the Run Configurations, for the Proxy Project	If everything has been entered **Successfully**, the following should be written to the **ECLIPSE IDE Console Window:** **Reference the 2nd Line from the Bottom - It states that the TomCat Server has started on the default port(s) 8080** ``` /_/_/_/ /_/ /_/ /_/ /_/_/_/ /_/ /_/ /_/ /_/ /_/_/ /_/ /_/ /_/ /_/ /_/ /_/ /_/ /_/ /_/ /_/ /_/ /_/ /_/ /_/ /_/ /_/ /_/ /_/ /_/_/ /_/_/ /_/ /_/_/ /_/_/ /_/ /_/ /_/_/ /_/ /_/ /_/ /_/ Proxy: (v2.0.0.RELEASE) Spring Boot: (v2.2.6.RELEASE) 2020-06-11 13:03:41.886 INFO 20320 --- [main] c.a.a.docmgmt.proxy.FrontendApplication : The following profiles are active: local 2020-06-11 13:03:43.018 WARN 20320 --- [main] o.s.boot.actuate.endpoint.EndpointId : Endpoint ID 'hystrix.stream' contains invalid characters, please migrate t 2020-06-11 13:03:43.148 INFO 20320 --- [main] o.s.cloud.context.scope.GenericScope : BeanFactory id=0ebb10df-185f-3481-9bc3-0a77d6dc9a7f 2020-06-11 13:03:43.324 INFO 20320 --- [main] trationDelegate$BeanPostProcessorChecker : Bean 'com.compozed.appfabric.logging.config.AppFabricLoggingConfiguration' 2020-06-11 13:03:43.479 INFO 20320 --- [main] trationDelegate$BeanPostProcessorChecker : Bean 'org.springframework.cloud.autoconfigure.ConfigurationPropertiesRebin 2020-06-11 13:03:43.784 INFO 20320 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port(s): 8080 (http) 2020-06-11 13:03:43.799 INFO 20320 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat] 2020-06-11 13:03:43.799 INFO 20320 --- [main] org.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/9.0.33] 2020-06-11 13:03:44.044 INFO 20320 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext 2020-06-11 13:03:44.045 INFO 20320 --- [main] o.s.web.context.ContextLoader : Root WebApplicationContext: initialization completed in 2129 ms 2020-06-11 13:03:44.199 WARN 20320 --- [main] c.n.c.sources.URLConfigurationSource : No URLs will be polled as dynamic configuration sources. 2020-06-11 13:03:44.199 INFO 20320 --- [main] c.n.c.sources.URLConfigurationSource : To enable URLs as dynamic configuration sources, define System property ar 2020-06-11 13:03:44.217 INFO 20320 --- [main] c.netflix.config.DynamicPropertyFactory : DynamicPropertyFactory is initialized with configuration sources: com.netf 2020-06-11 13:03:45.621 INFO 20320 --- [main] o.s.cloud.commons.util.InetUtils : Cannot determine local hostname 2020-06-11 13:03:46.028 WARN 20320 --- [main] o.s.c.n.a.ArchaiusAutoConfiguration : No spring.application.name found, defaulting to 'application' 2020-06-11 13:03:46.028 WARN 20320 --- [main] c.n.c.sources.URLConfigurationSource : No URLs will be polled as dynamic configuration sources. 2020-06-11 13:03:46.028 INFO 20320 --- [main] c.n.c.sources.URLConfigurationSource : To enable URLs as dynamic configuration sources, define System property ar 2020-06-11 13:03:46.213 INFO 20320 --- [main] o.s.s.concurrent.ThreadPoolTaskExecutor : Initializing ExecutorService 'applicationTaskExecutor' 2020-06-11 13:03:46.795 INFO 20320 --- [main] o.s.c.n.zuul.ZuulFilterInitializer : Starting filter initializer 2020-06-11 13:03:46.804 INFO 20320 --- [main] o.s.b.a.e.web.EndpointLinksResolver : Exposing 2 endpoint(s) beneath base path '/actuator' 2020-06-11 17:03:46,872[appFabric-logging]enable|2.6.2| 2020-06-11 17:03:46,873[appFabric-starter-proxy]enable|2.6.2| 2020-06-11 13:03:46.900 INFO 20320 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port(s): 8080 (http) with context path '' 2020-06-11 13:03:48.206 INFO 20320 --- [main] o.s.cloud.commons.util.InetUtils : Cannot determine local hostname 2020-06-11 13:03:48.210 INFO 20320 --- [main] c.a.a.docmgmt.proxy.FrontendApplication : Started FrontendApplication in 10.332 seconds (JVM running for 13.288) 2020-06-11 13:05:51.093 INFO 20320 --- [n(11)-127.0.0.1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet' 2020-06-11 13:05:51.094 INFO 20320 --- [n(11)-127.0.0.1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet' 2020-06-11 13:05:51.102 INFO 20320 --- [n(11)-127.0.0.1] o.s.web.servlet.DispatcherServlet : Completed initialization in 7 ms ``` **As before, this can also be confirmed from the Boot Dashboard Perspective (Reference Snapshot below...)** Console Boot Dashboard Problems Javadoc Declaration Search RestClient Tool Gradle Tasks Gradle Executions Error Log local Install local cloud services ab-docmgmt-proxy [:8080] ab-docmgmt-service [:8081]
15	Right-Click on the Desktop Shortcut **Co mmand Prompt** and " **Run Elevated with DefendPoint**" ***It is a always good practice to run applications as "**Run Elevated with DefendPoint**"	Should open a Command Prompt running with Elevated Rights.

16	Navigate to the Proxy Project	Administrator: c:\windows\system32\cmd.exe
		`C:\workspace_edmapis>cd *proxy`
		
		`C:\workspace_edmapis\ab-docmgmt-proxy>dir`
		`Volume in drive C is Allstate1`
		`Volume Serial Number is 4AAD-1E9C`
		
		`Directory of C:\workspace_edmapis\ab-docmgmt-proxy`
		
		`06/10/2020 02:21 PM <DIR> .`
		`06/10/2020 02:21 PM <DIR> ..`
		`06/10/2020 04:40 PM 993 .classpath`
		`06/03/2020 03:25 PM 1,640 .gitignore`
		`06/09/2020 04:59 PM <DIR> .gradle`
		`06/10/2020 05:27 PM 793 .project`
		`06/11/2020 01:21 PM <DIR> .settings`
		`06/03/2020 03:25 PM 4,870 BACKUP_build.gradle`
		`06/10/2020 04:40 PM <DIR> bin`
		`06/09/2020 05:00 PM <DIR> build`
		`06/09/2020 04:58 PM 5,008 build.gradle`
		`06/03/2020 03:25 PM <DIR> gradle`
		`06/03/2020 03:25 PM 5,471 gradlew`
		`06/03/2020 03:25 PM 2,260 gradlew.bat`
		`06/03/2020 03:25 PM 8,030 jenkinsFile.groovy`
		`06/03/2020 03:25 PM 2,683 jenkinsFileProd.groovy`
		`06/03/2020 03:25 PM 772 jenkinsFilePullRequest.groovy`
		`06/10/2020 02:21 PM <DIR> lib`
		`06/09/2020 05:00 PM 104 logErrors.txt`
		`06/09/2020 05:00 PM 5,136,335 logStatements.txt`
		`06/03/2020 03:25 PM 392,445 pipeline-frontend.png`
		`06/03/2020 03:25 PM 433,399 pipeline.png`
		`06/03/2020 03:25 PM 7,879 README.md`
		`06/03/2020 03:25 PM 39 settings.gradle`
		`06/03/2020 03:25 PM <DIR> src`
		`16 File(s) 6,002,721 bytes`
		`9 Dir(s) 92,163,051,520 bytes free`
		
		`C:\workspace_edmapis\ab-docmgmt-proxy>`

17	Execute the following **Mock ISAM** at the command line	Enter the following **GradleW** Statement and **MASH** the **ENTER KEY**
		
		a. gradlew -debug StartMockIsam > mockISAMLog.txt 2> mockISAMErrorLog.txt
		
		This will Start the **Mock ISAM Application on port 7000**
		
		`C:\workspace_edmapis\ab-docmgmt-proxy>gradlew StartMockIsam`
		`Picked up _JAVA_OPTIONS: -Xmx6144m`
		
		`> Task :StartMockIsam`
		
		`BUILD SUCCESSFUL`
		
		`C:\workspace_edmapis\ab-docmgmt-proxy>`

18	Within a Browser Window, navigate to the following **Mock ISAM** URL	**`MockISAM URL:`** `http://localhost:7000` You should see the following:

User ID

Password

Groups

Login

This is a private computer facility. Access to it for any reason must specifically be authorized. Unless you are specifically authorized, your continued access and further inquiry will expose you to prosecution to the fullest extent of federal, state and local laws.

By logging on and continuing further on this site, the user agrees that he/she will not, at any time, or in any manner, directly or indirectly, disclose to any third party or permit any third party to access any confidential information concerning matters affecting or relating to the pursuits of Allstate, except upon direct written authority of Allstate. The user agrees that he/she will not use, or permit to be used, Allstate property or any information obtained or derived therefrom in any manner, directly or indirectly, other than in the performance of services for Allstate.

By default, AppFabric is configured to use the AppFabric ISAM provider. The ISAM provider requires request header
frontend proxy includes **Mock ISAM**. **Mock ISAM** is a mock proxy that mimics ISAM. Mock ISAM can also be useful f

If the **MockISAM** does not spin up **OR** is not found, most likely **Step #17** was **NOT Successful After-all**. Review the

For example: When I attempted to spin up the **MockISAM** Instance, the error logs reported the following:

13:42:23.880 [ERROR] [system.err] **Error: Invalid or corrupt jarfile** C:\workspace_edmapis\ab-docmgmt-proxy/lib/c

13:42:23.916 [ERROR] [org.gradle.api.Task] Unable to start the mockisam jar
org.gradle.process.internal.ExecException: Process 'command 'C:\Program Files\AdoptOpenJDK\jdk-8.0.232.09-hots

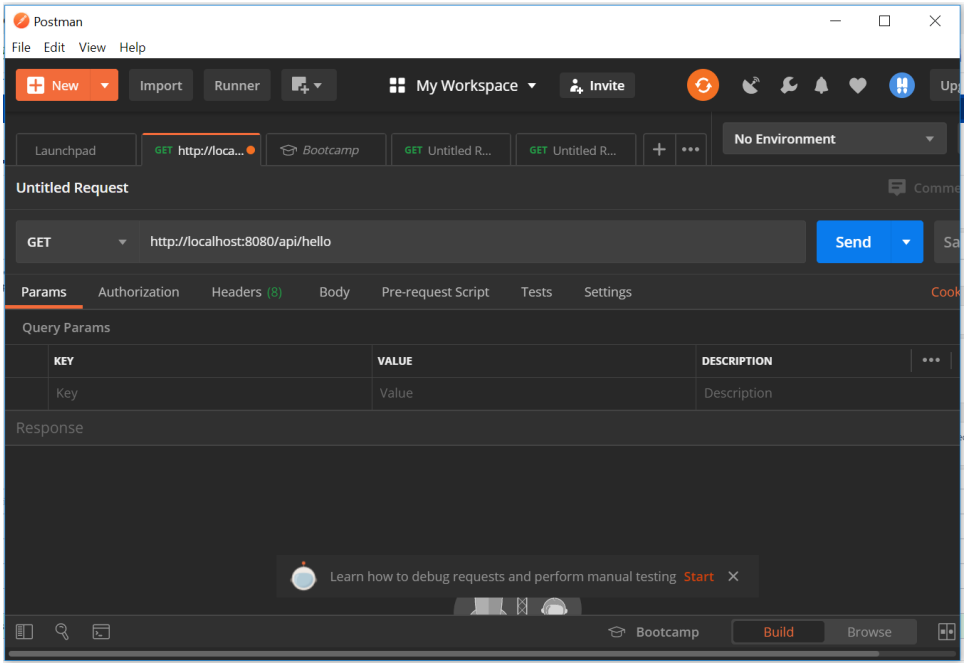
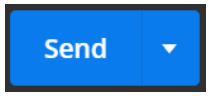
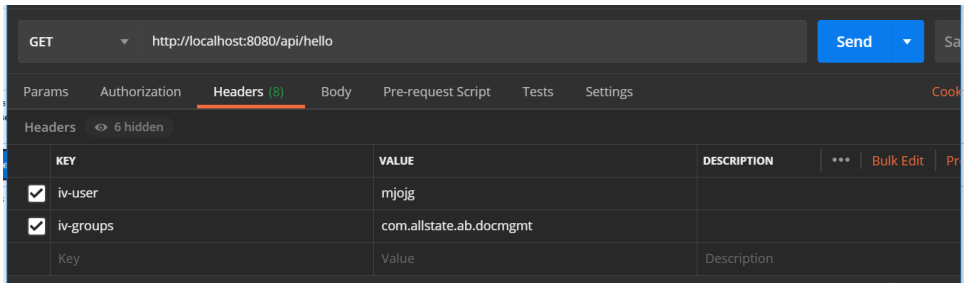
My solution was to manually retrieve it from the Artifactory and place it at the Classpath of the Proxy Project (Referer

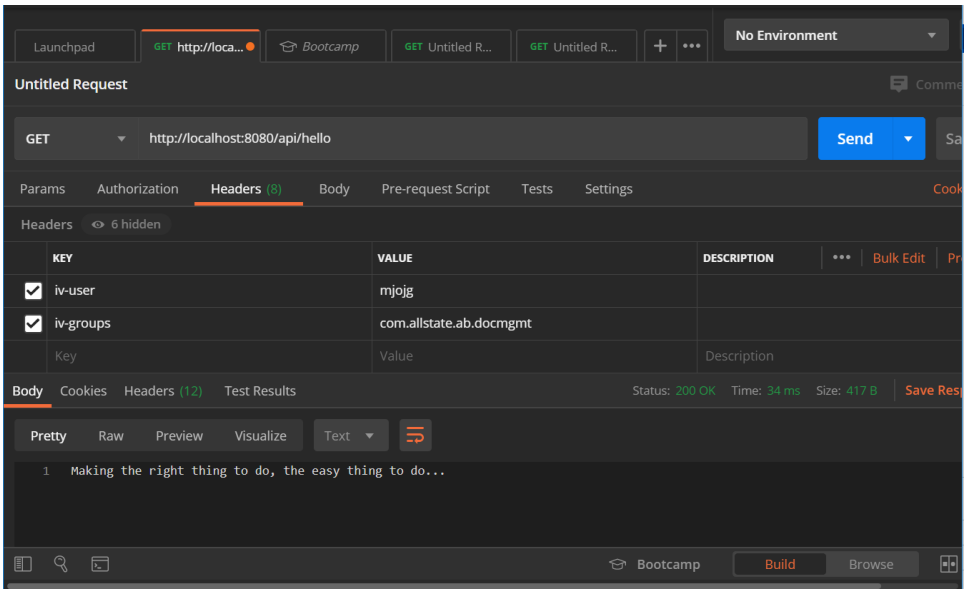
https://artifactory.allstate.com/artifactory/AITI-Engineering-Infrastructure-and-Experience-release/com/compozed/app

Then i navigated to the Proxy Lib Folder and manually ran the Jar File as shown below

```
C:\workspace_edmapis\ab-docmgmt-proxy\lib>dir
 Volume in drive C is Allstate1
 Volume Serial Number is 4AAD-1E9C

 Directory of C:\workspace_edmapis\ab-docmgmt-proxy\lib

06/11/2020  02:38 PM    <DIR>          .
06/11/2020  02:38 PM    <DIR>          ..
06/11/2020  02:14 PM        41,534,598 com-compozed-appfabric-mockisam-2.7.0.jar
06/11/2020  02:17 PM             1,826 com-compozed-mockisam-latest.jar
06/11/2020  02:14 PM    <DIR>          _REFERENCE
               2 File(s)     41,536,424 bytes
               3 Dir(s)  92,125,147,136 bytes free

C:\workspace_edmapis\ab-docmgmt-proxy\lib>java -jar com-compozed-appfabric-mockisam-2.7.0.jar
Picked up _JAVA_OPTIONS: -Xmx6144m

  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::        (v2.3.0.RELEASE)

2020-06-11 14:39:05.531  INFO 11480 --- [           main] c.c.a.mockisam.IsamFakeApplication       : No
 active profile set, falling back to default profiles: default
2020-06-11 14:39:08.546  WARN 11480 --- [           main] o.s.boot.actuate.endpoint.EndpointId     : En
dpoint ID 'hystrix.stream' contains invalid characters, please migrate to a valid format.
2020-06-11 14:39:08.788  INFO 11480 --- [           main] o.s.cloud.context.scope.GenericScope     : Be
anFactory id=36b8c260-e690-3187-a4c5-ddb191a87a42
2020-06-11 14:39:10.150  INFO 11480 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : To
mcat initialized with port(s): 7000 (http)
2020-06-11 14:39:10.176  INFO 11480 --- [           main] o.apache.catalina.core.StandardService   : St
arting service [Tomcat]
2020-06-11 14:39:10.176  INFO 11480 --- [           main] org.apache.catalina.core.StandardEngine  : St
arting Servlet engine: [Apache Tomcat/9.0.35]
2020-06-11 14:39:10.404  INFO 11480 --- [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       : In
itializing Spring embedded WebApplicationContext
2020-06-11 14:39:10.404  INFO 11480 --- [           main] o.s.web.context.ContextLoader            : Ro
ot WebApplicationContext: initialization completed in 4801 ms
2020-06-11 14:39:10.570  WARN 11480 --- [           main] c.n.c.sources.URLConfigurationSource     : No
```

19	If you have not already done so, Download the "Postman" Application.	This will require "Elevated Rights", as described throughout this learning curve, in order to download/install an applica Navigate to the following URL below and download the "**Postman**" Application for a 64bit Operating System. https://www.postman.com/downloads/ This application will allow the developer a means of testing/debugging their Service/Proxy/MockISAM Applications th APPFabric CLI Framework Command line.
20	Open the Postman Application	Should open and present a welcome screen

21	Test the API	Enter the following **API GET REQUEST** as shown in the screen-shot below: **a.** http//localhost:8080/api/hello 
22	Navigate to the **Headers** Tab and enter **KEY/VALUE** pairs for '**iv-user**' & '**iv-groups**'	The **KEY/VALUE** pairs for '**iv-user**' & '**iv-groups**' will be specific to your Network Credentials and the group established for yo (Reference the ***build.gradle*** file for groups as an example)
23	Click the Send Button 	Will submit the REQUEST the the Proxy on the TOMCAT Server 

24	Validate the RESPONSE has been received	The developer should see the following message in the **RESPONSE** Body "***Making the right thing to do, the easy thing to*** 
25	Congratulations, you have successfully Run an APPFabric Service & Proxy Application...	